10.6

16. $F = \frac{s_1^2}{s_2^2} = \frac{114^2}{103^2} = 1.225$

$F_{14,14,0.05/2} = 2.9786$

$H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 \neq \sigma_2^2$

$\because 1.225 < 2.9786$

$\therefore$ Non-reject H_0

p-value $= P(F > 1.225 \mid 14, 14) \approx 0.71$

10.6

14. $H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 \neq \sigma_2^2$

$F = \frac{s_2^2}{s_1^2} = 3.268$

p-value $= P(F > 3.268 \mid 29, 29)$

22. $H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 \neq \sigma_2^2$

$F = \frac{s_2^2}{s_1^2} = \frac{2.96}{1.54} = 1.922$

$P(F \geq 1.922 \mid 14, 14) =$

10.) (Review)

11. $\mu_1 = 1.875$

$S_1 = 0.835$

$\mu_2 = 3$

$S_2 = 0.756$

$H_0: \mu_1 = \mu_2$

$H_a: \mu_1 \neq \mu_2$ $\quad \alpha = 0.05$

$F = \frac{S_1^2}{S_2^2} = 1.2199$

$F_{7,7,0.025} = 4.9949$

$$T = \frac{\mu_1 - \mu_2}{\sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} = \frac{-1.125}{\sqrt{2.044\left(\frac{1}{8}+\frac{1}{8}\right)}} = \frac{-1.125}{\sqrt{\frac{4.088}{8}}} = -1.577$$

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} = \frac{\cancel{7}\cdot 1.875^2 + \cancel{7}\cdot 0.756^2}{\cancel{14}\ 2} = 2.044$$

$T_{14,0.025} = 2.3646$

$\because |-1.577| < 2.3646$

$\therefore$ Non-reject H_0

12. $d_{1-2} = 1, -1, -2, 1, -1, 1, 0, -1$

$\bar{d} = -0.75$

$S_d = 0.886$

$H_0: \mu_d = 0$

$H_a: \mu_d \neq 0$

$t = \frac{-0.75}{0.886/\sqrt{8}} = -2.392$

$t_{7, 0.025} = 2.3646$

$\therefore 2.392 > 2.3646$

$\therefore$ Reject H_0